NYCU Research Overview

Mar. 21, 2024

Integrated Circuits and Smart Systems Lab

PI: Yi-Chung Wu 吳易忠 (yicwu@nycu.edu.tw)

Institute of Electronics, National Yang Ming Chiao Tung University

ICSSL

Principle Investigator – Yi-Chung Wu

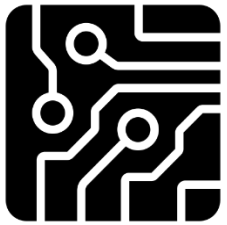
- **專長**
  - **數位IC與SoC系統設計**, 前瞻應用演算法與硬體開發
- **學歷**
  - 台灣大學電子研究所博士, 2021
  - 交通大學電子研究所碩士, 2016
  - 交通大學電機資訊學士班, 2013
- **經歷**
  - 台灣大學電子所 博士後研究, 2021.04-2024.02
  - 矽晶定序高速科技 電路設計, 2020.08-2023.12
  - 美國密西根大學 千里馬計畫訪問學者, 2019.09-2020.05
  - 揚智科技Ali Tech. 產業贊助實習生, 2017.09-2018.08
  - 美國伊利諾大學香檳校區 交換學生, 2012.08-2013.01

Awards and Honors

- **2024**
  - 國科會度博士後研究人員學術研究獎
- **2023**
  - TSIA半導體獎：具博士學位之新進研究人員
  - VLSI Design CAD大會最佳論文獎
  - TSRI優良晶片特別設計獎
- **2022**
  - ASSCC Best Student Design Award
  - IEEE台北分會博士論文獎
  - 台大電子所學生最佳博士論文獎
  - 台灣生醫電子工程協會最佳博士論文優等獎
- **2021**
  - 國研院研發服務平台亮點成果獎優等獎
  - 台灣積體電路設計學會博士論文獎
  - 台灣半導體研究中心優良晶片特優設計獎
- **2020**
  - ISSCC Takuo Sugano Award for Outstanding Far-East Papers
  - 台大電子所學生傑出研究獎
  - 旺宏金矽獎金獎
  - 戴樂格半導體獎學金
  - 台灣半導體研究中心優良晶片特別設計獎
  - 臺大1975級電機系系友科技研究創新獎
- **2019**
  - IEEE SSCS Predoctoral Achievement Award
  - 旺宏金矽獎銅獎
- **2018**
  - 聯詠科技博士生獎學金
  - TSIA半導體獎：博士研究生
- **2017**
  - ISSCC Silkroad Award
  - 國家晶片系統設計中心優良晶片特別設計獎
  - 台灣生醫電子工程協會最佳碩士論文優等獎
  - 台大電子所學生傑出研究獎
  - 旺宏金矽獎銅獎
  - 教育部 IC/CAD 競賽研究所全客戶設計組完成獎
  - 中國電機工程學會青年論文獎第二名
- **2015**
  - 全國大學校院智慧電子系統設計競賽核心技術組特優
  - 教育部 IC/CAD 競賽標準元件數位電路設計組特優

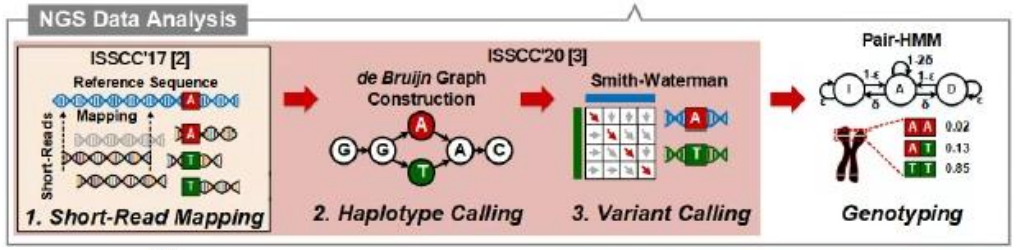
Prerequisites Courses and EE Projects

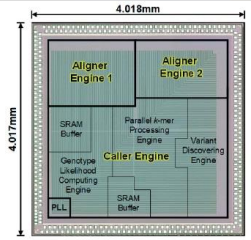
- **大學部**
  - 計算機概論與程式設計
  - 邏輯設計
  - **數位電路與系統**
  - 超大型積體電路導論
  - VLSI實驗
- **研究所**
  - **積體電路設計實驗**
  - 數位積體電路
- **專題訓練 - 數位IC Flow**
  - 探索實際應用
  - 論文閱讀訓練
  - 演算法開發
  - 電路架構設計
  - **數位IC設計流程**
  - FPGA驗證



Advanced Research Topics

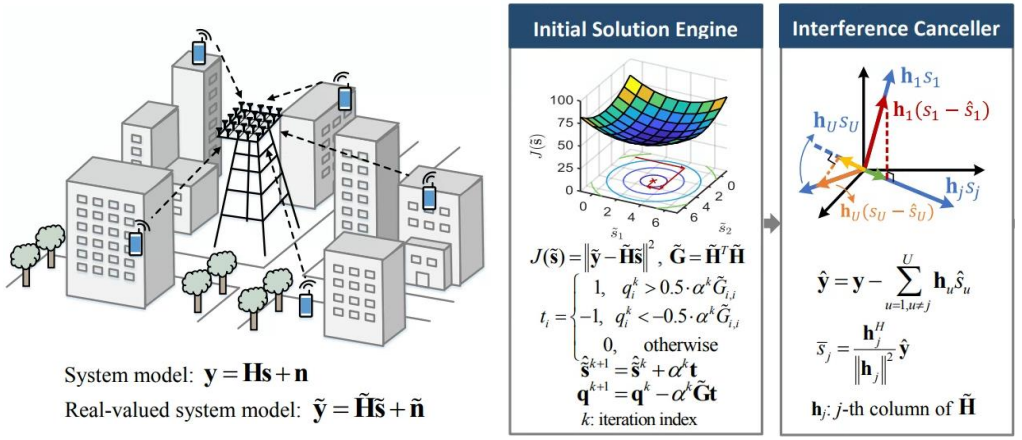
Biomedical Signal Processing



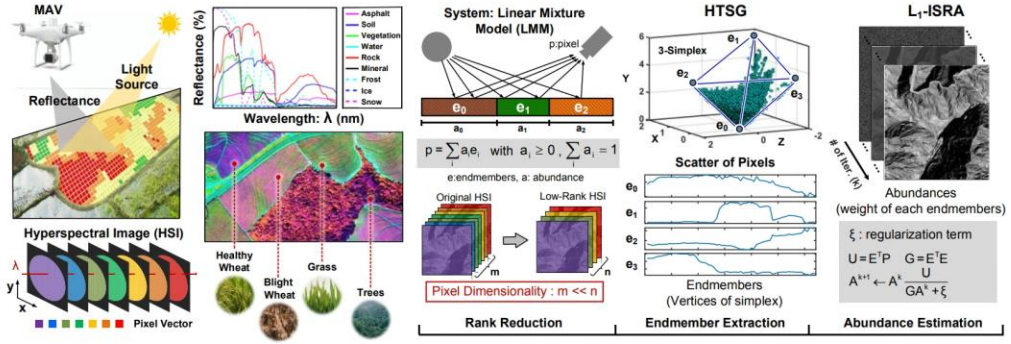


Technology		TSMC 28nm CMOS
Area	Chip	4.018 x 4.017 mm²
	Core	3.800 x 3.799 mm²
Gate Count		28.51 M
Core V_{DD}		0.9 V
I/O V_{DD}		1.8 V
Frequency		400MHz
Overall Power		2.73 W

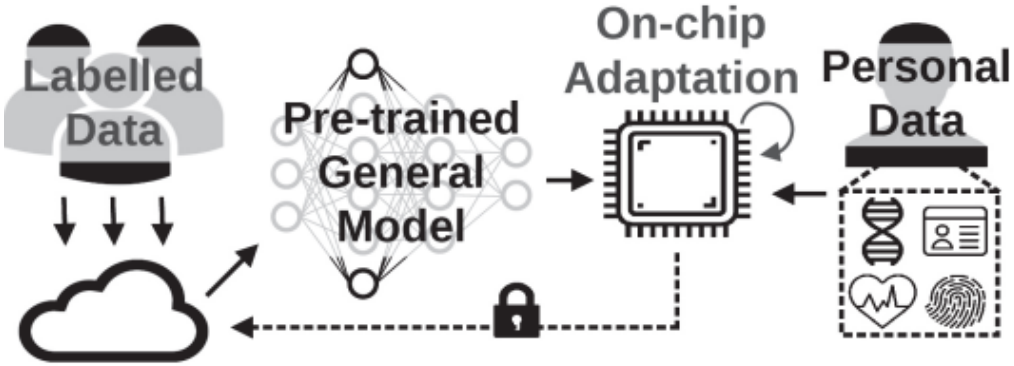
High-Performance Computing



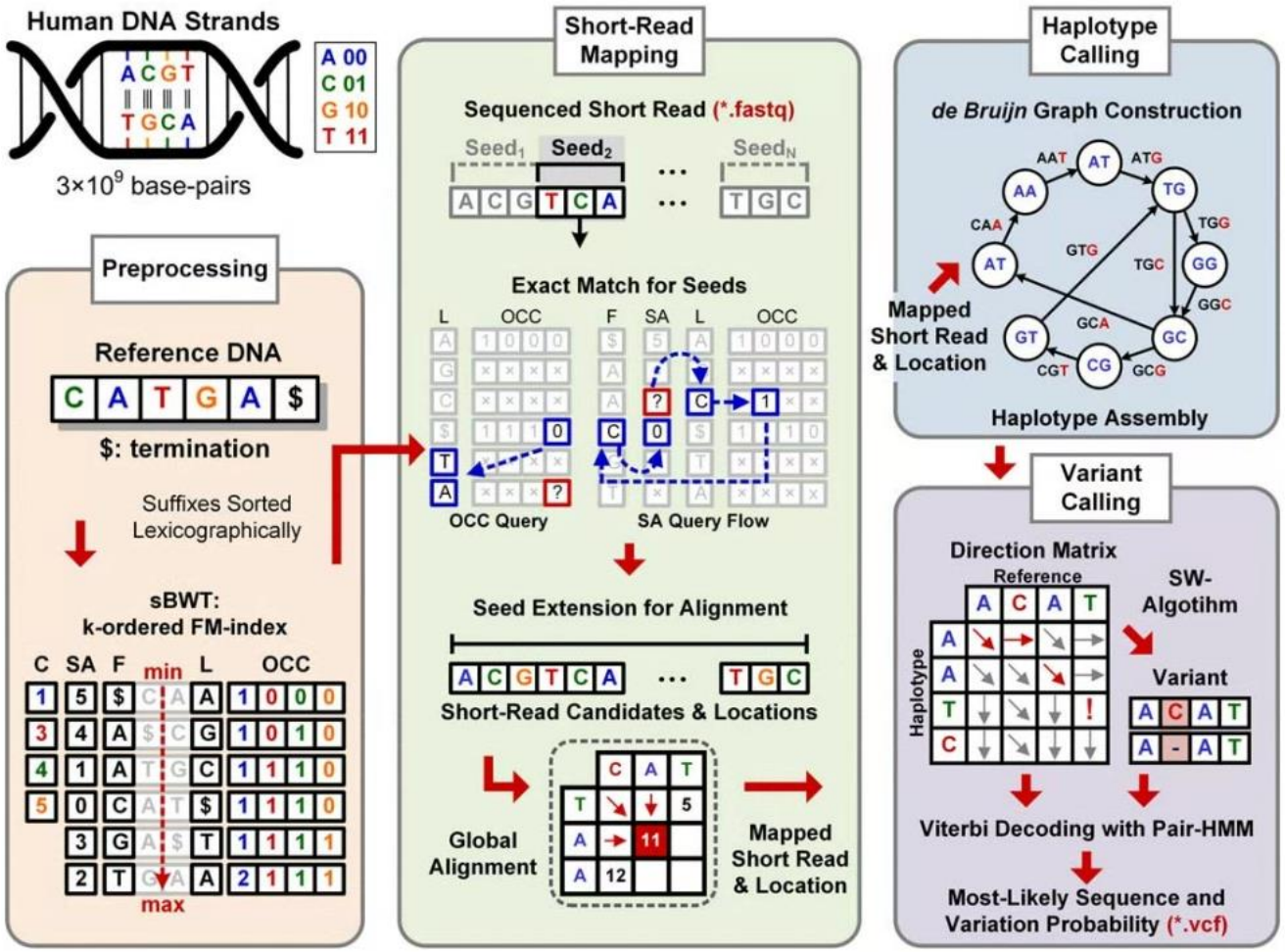
VLSI for Multi-Media

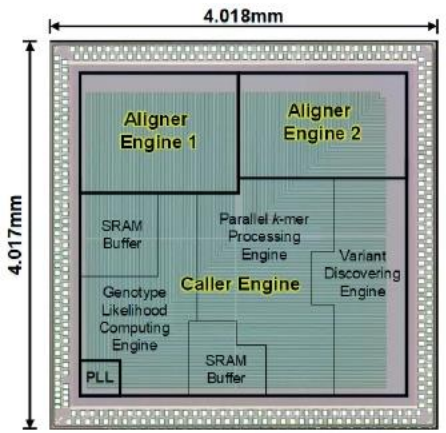


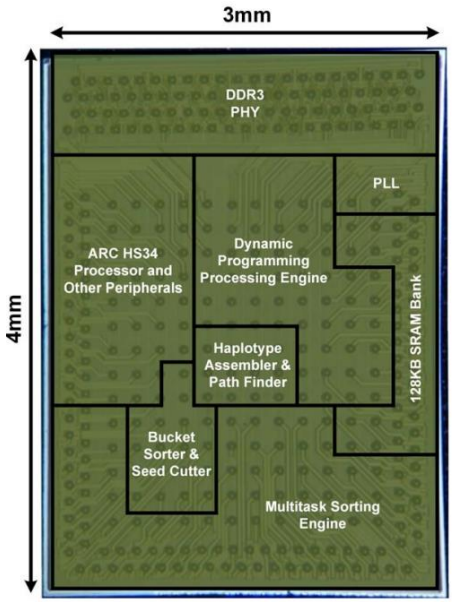
AI/ML Accelerator

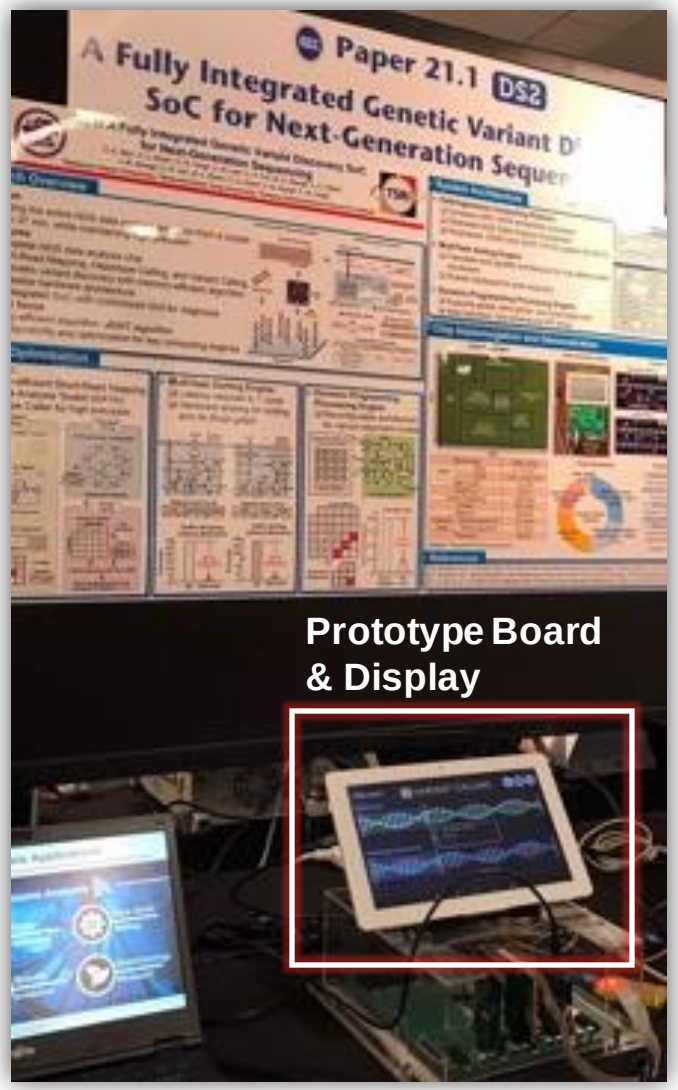


Research Interests: Biomedical Circuits Design

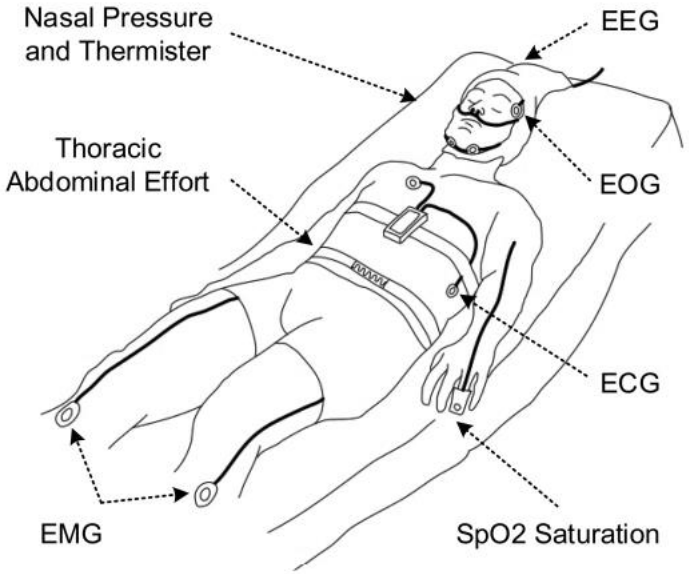


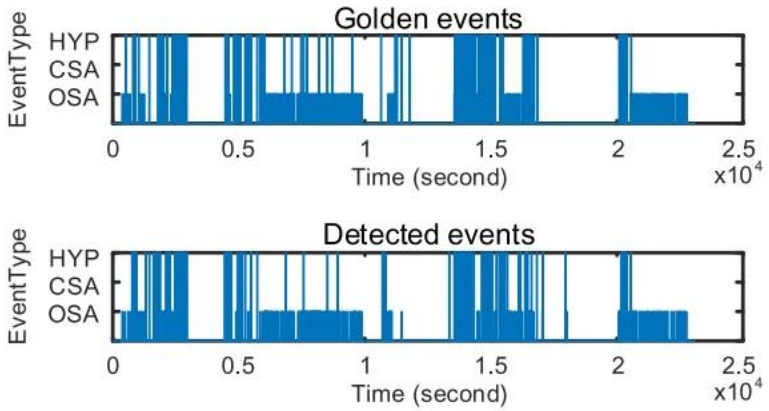


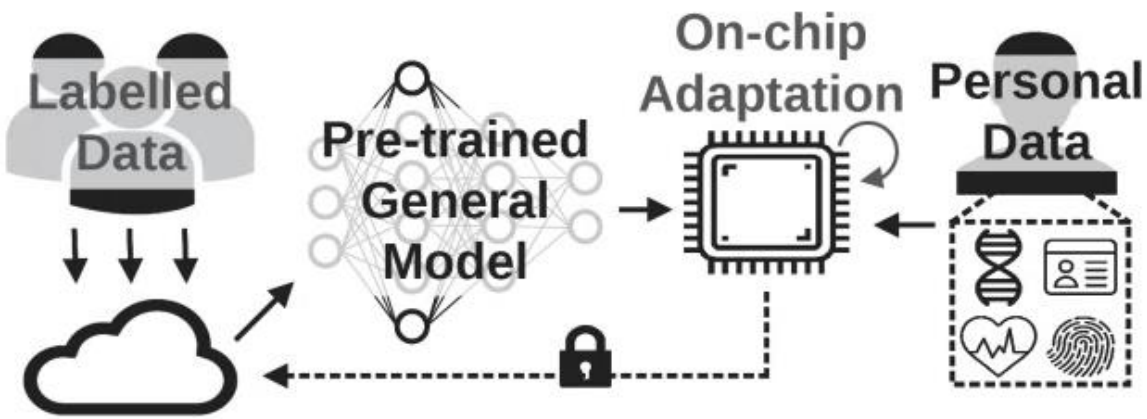


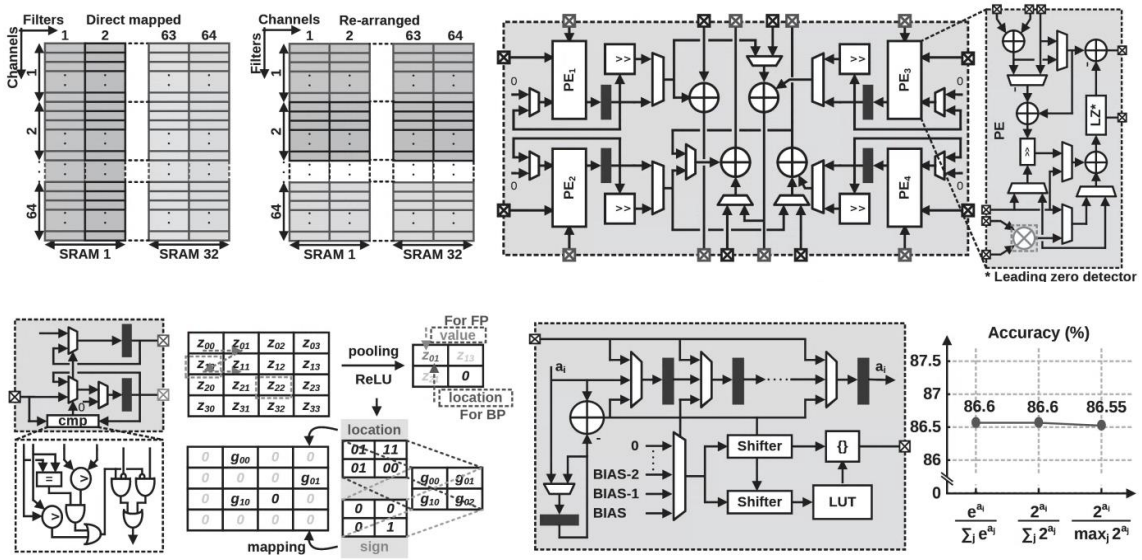


Research Interest: Machine Learning

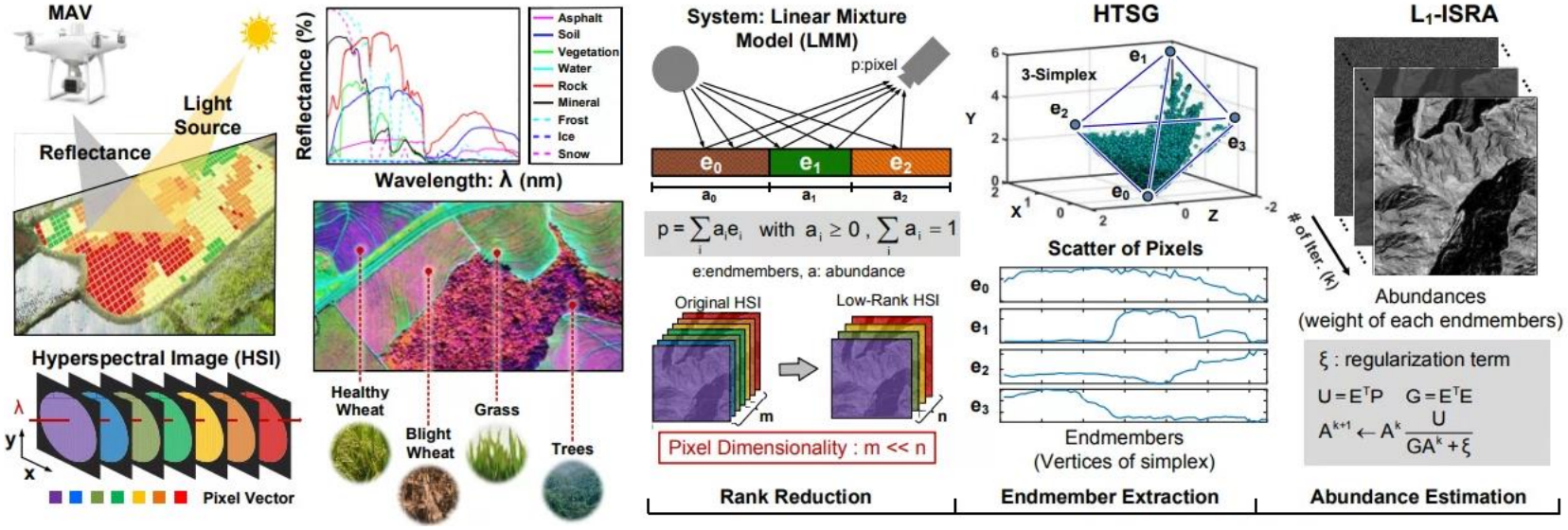


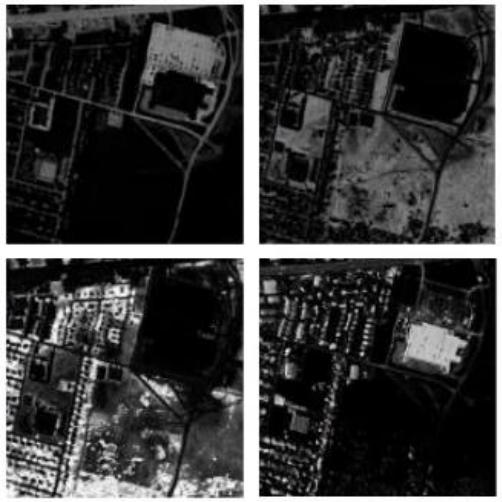




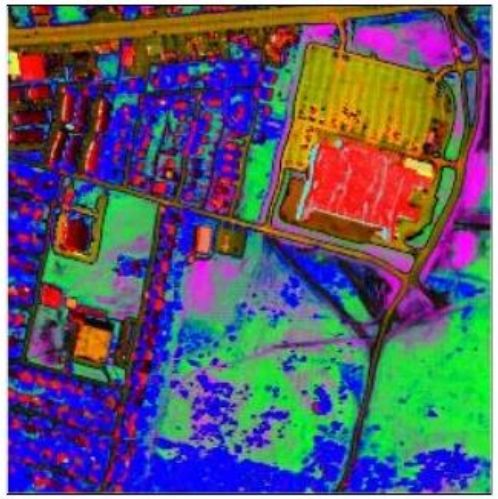


Research Interest: High-Performance Computing

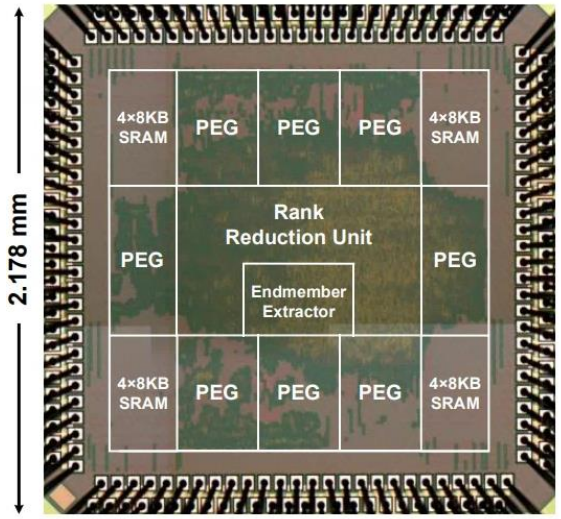




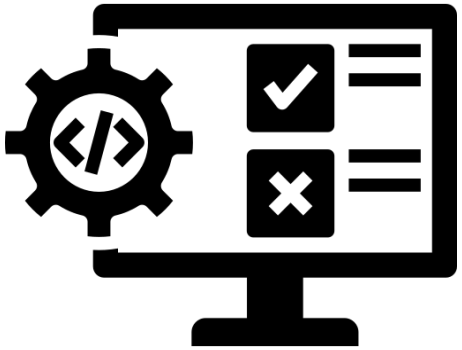
Four Selected Abundances



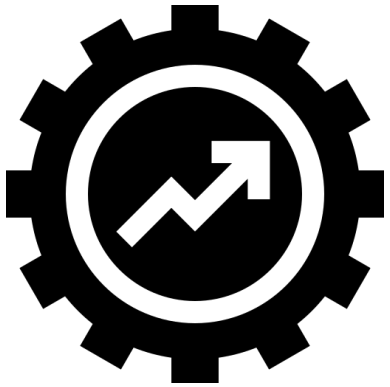
Fusion of abundances



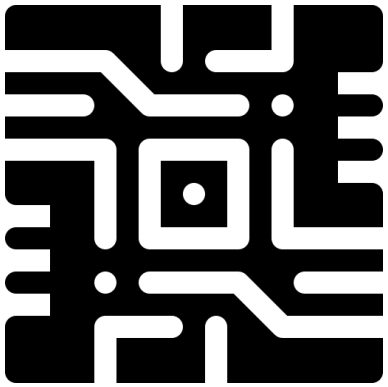
Goals and Prospects



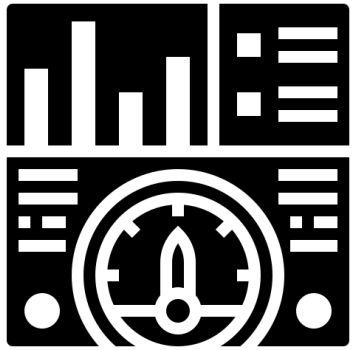
Software Prototype



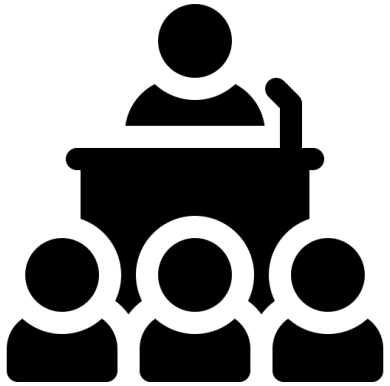
Algorithm Optimization



Chip and System Design



Measurement and Analysis



Conference Oral Presentation

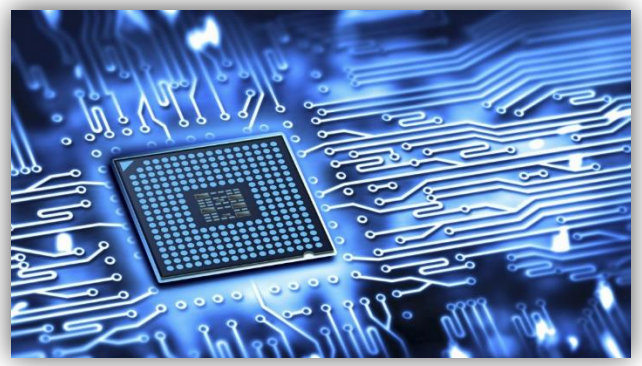
Journal Article and Thesis

Lab Advantages

新穎學術研究
- 跨領域研究: 生醫+電子
- 系統單晶片整合
- 數位系統與平台開發

團隊、國際機構交流
- 跨領域合作
- 大型整合計畫
- 國際合作: 美國

學術界、產業界合作
- 醫療院所、機構
- 大型成熟企業
- 潛在新創產業



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Welcome to Join Me

PI: Yi-Chung Wu (吳易忠)

https://sites.google.com/view/nycu-icssl

Feel free to email me to discuss
research topics, contests preparation, experiences, and more.

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